

14th Dec

Binomial Distribution,

$n \neq p$ $n \rightarrow$ Series of Independent Experiments similar to Bernoulli distribution but here we have a number of experiments and their probability.

- ① Discrete Random variable
- ② Every outcome of the Experiments is binary
- ③ These Experiments are performed for n trials

Eg: Tossing a coin 10 Times $n = 10$

$\{H, T\}$

Notation: $B(n, p)$

Parameters: $n \in \{0, 1, 2, \dots\} \Rightarrow$ n. of trials
or experiments

$p \in [0, 1] \Rightarrow$ success probability

$$q = 1 - p$$

Support $K \in \{0, 1, 2, 3, \dots, n\} \Rightarrow$ No. of Success

PMF

$$P_X(K, n, p) = {}^n C_K p^K (1-p)^{n-K}$$

for $K = 0, 1, 2, \dots, n$ where

Binomial Coefficient

$${}_n C_K = \frac{n!}{K! (n-K)!}$$

$$\left\{ \begin{array}{l} \text{Mean} = np \\ \text{Variance} = npq \\ \sigma = \sqrt{npq} \end{array} \right\}$$

Eg:- Coin flip

No. of trials (n) = 5

probability of success (p) = 0.5

No. of success (k) = Var. fr 0 to 5

Q What is the probability of getting exactly 3 heads in 5 flips

$$n=5 \quad k=3$$

$$P(X=3) = {}^5C_3 (0.5)^3 (1-0.5)^{5-3}$$

$$= 0.3125$$